CSCI-4041
Spring 2026
Paper Homework 3 - Chapters 7 and 10

Name: Vector Ekanmadjanes **Section:** _____

1. Using Figure 7.1 as a model, illustrate the operation of PARTITION on the array $A = [13, 19, 9, 5, 12, 8, 7, 4, 21, 2, 6, 11]$.

[13, 19, 9, 5, 12, 8, 7, 4, 21, 2, 6, 11]
pivot

↓ 9 < 11

[9, 19, 13, 5, 12, 8, 7, 4, 21, 2, 6, 11]

↓ 5 < 11

[9, 5, 13, 19, 12, 8, 7, 4, 21, 2, 6, 11]

↓ 8 < 11

[9, 5, 8, 7, 12, 13, 11, 4, 21, 2, 6, 11]

↓ 4 < 11

[9, 5, 8, 7, 4, 13, 19, 12, 21, 2, 6, 11]

↓ 2 < 11

[9, 5, 8, 7, 4, 2, 19, 12, 21, 13, 6, 11]

↓ 6 < 11

[9, 5, 8, 7, 4, 2, 6, 12, 21, 13, 19, 11]

↓

[9, 5, 8, 7, 4, 2, 6, 11, 21, 13, 19, 12]
pivot

2. Consider applying the PARTITION method to a sub-array with all the same value.

(a) What value of q does PARTITION return when all elements in the sub-array $A[p : r]$ have the same value? You should write a sentence or two explaining your logic and arguments.

- i get incremented everytime as $j \leq j$ is true
- pivot at last element
- $q = r$, worst case for quicksort

(b) Modify PARTITION so that $q = \lfloor (p + r)/2 \rfloor$ when all elements in the sub-array $A[p : r]$ have the same value.

- flag can be added so that element does not end up on one side

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if A[j] == x:
    if flag == true:
        i += 1
        then swap A[i] and A[j]
        flag = !flag
```

3. Show that RANDOMIZED-QUICKSORT's expected running time is $\Omega(n \log n)$.

$-\Omega(n \log n)$

$- \frac{2}{j-i+1}$

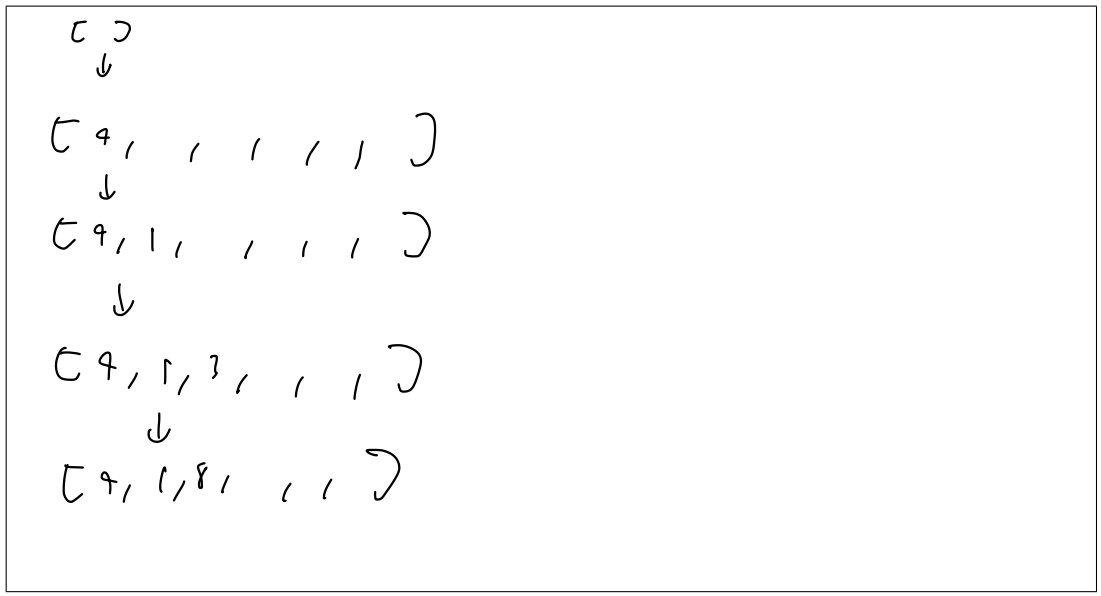
$k = j - i$

$= \sum_{k=1}^{n-i} \frac{1}{k+1} > \sum_{k=1}^{n/2} \frac{1}{k+1}$

$E(x) \geq 2 \cdot (\frac{n}{2}) \Omega(\log n) = \Omega(n \log n)$

4. Illustrate the result of the following operations given they are implemented in the array-based version in the textbook.

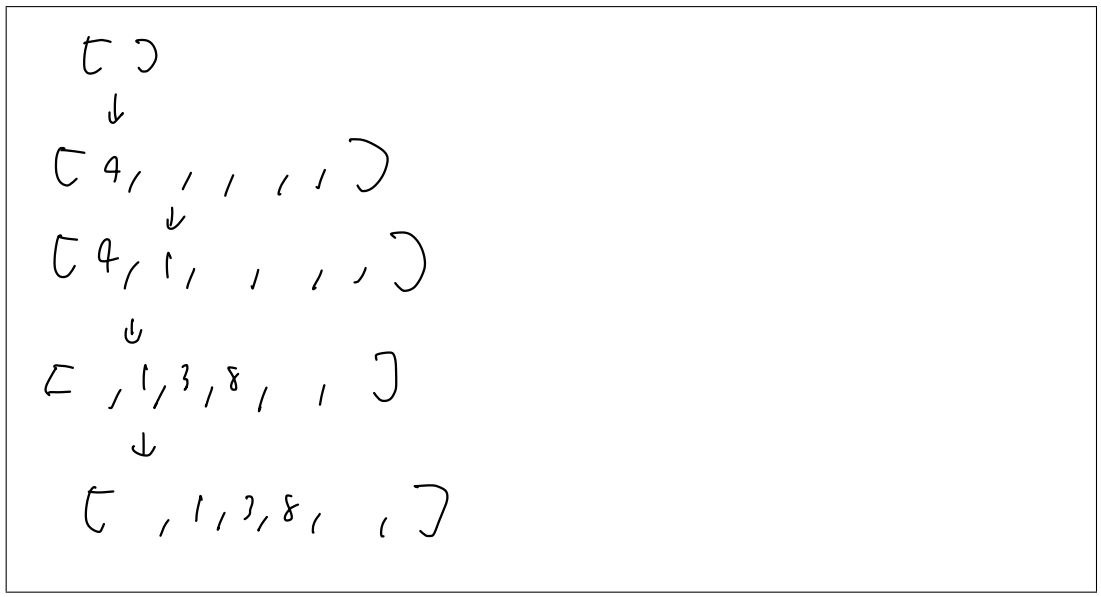
   (a) Using Figure 10.2 as a model, illustrate the result of each operation in the sequence $PUSH(S, 4), PUSH(S, 1), PUSH(S, 3), POP(S), PUSH(S, 8)$, and $POP(S)$ on an initially empty stack S stored in array $S[1 : 6]$.



   (b) Using Figure 10.3 as a model, illustrate the result of each operation in the following sequence:

$$ENQUEUE(Q, 4), ENQUEUE(Q, 1), ENQUEUE(Q, 3),$$
$$DEQUEUE(Q), ENQUEUE(Q, 8), \text{ and } DEQUEUE(Q),$$

   on an initially empty queue Q stored in array $Q[1 : 6]$.



5. Consider a singly linked list such as Joy's example given in the Canvas module.

(a) Explain why the dynamic-set operation INSERT on a singly linked list can be implemented in O(1) time.

- node can be inserted at the start of the list
- no traversal needed
- new node next can point to head
- new node become head

(b) Explain why the worst-case time for DELETE is $\Theta(n)$.

- must traverse list once → O(n) to find prev node and link to the current.next node